

Positive Square Energy of Every Pentacyclic Cactus

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Abstract

Let G be a connected pentacyclic cactus of order n . We prove $s^+(G) > n$. The exact block-additive doubly nonnegative estimate reduces the problem to the cycle multisets $\{3, 3, 3, 3, q\}$ and $\{3, 3, 3, 5, 5\}$. For a disconnected shared-cut graph, we give the complete lists of eleven and fifteen colored cluster partitions. The two partitions not settled by direct packet addition require entry-sensitive interval decompositions of a central pentagon; the all-singleton partitions require reduced-tree leaf arguments. For one shared-cut cluster, the first residual is proved by opening the q -cycle or partitioning it among four triangular petals. The second is settled by an exact census of all forty colored incidence trees: thirty-six admit a positive one-cycle split, and four explicit exceptions are treated directly. Every construction allows arbitrary connector trees and arbitrary trees attached at arbitrary vertices.

1 Preliminaries and dependencies

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles. Its cyclomatic number is the number of cyclic blocks. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For every vertex partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_k$, induced-subgraph superadditivity gives

$$s^+(G) \geq \sum_{i=1}^k s^+(G[V_i]). \tag{1}$$

This is the theorem of Akbari, Kumar, Mohar, and Pragada [1].

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{2}, \quad \delta_q = \sec(\pi/q) - 1 < 1 \quad (q \equiv 1 \pmod{4}).$$

We use the following established packet bounds. All permit arbitrary trees attached at arbitrary

vertices; mixed packets may have arbitrary bridge connectors.

$$\sigma(T) > 0, \quad \sigma(P) \geq -\delta, \quad \sigma(C_q) \geq -\delta_q, \quad (2)$$

$$\sigma(TT) > 1, \quad \sigma(TP) > 1 - \delta, \quad \sigma(TC_q) > 1 - \delta_q, \quad (3)$$

$$\sigma(H) \geq 0 \quad \text{for every bicyclic or tricyclic cactus } H, \quad (4)$$

$$\sigma(TTT) > 2 \quad \text{for one shared-cut triangular cluster}, \quad (5)$$

$$\sigma(TTTT) > 3 \quad \text{for one shared-cut four-triangle cluster}, \quad (6)$$

$$\sigma(TTC_q) > 2 - \delta_q \quad \text{if the two triangles share a cut and } q \equiv 1 \pmod{4}, \quad (7)$$

$$\sigma(TPP) > 6 - 2\sqrt{5} \quad \text{for one shared-cut cluster}. \quad (8)$$

Also, every connected tetracyclic cactus has positive surplus:

$$\sigma(H) > 0. \quad (9)$$

Bounds (2)–(4) are proved in the bicyclic and tricyclic companion manuscripts [3, 4]. The shared TTC_q phase comparison (7) is the intersecting-triangle case of the $\{3, 3, q\}$ theorem in [4]. If every cycle is $3 \pmod{4}$ and the cycle-packing number is at most two, the Sachs half-plane theorem [7] gives

$$\sigma(H) > r - 1 \quad (10)$$

for a connected r -cyclic cactus. This proves (5). For (6), packing at most two again uses (10); the sole packing-three shared-cut incidence is a central triangle with three pairwise disjoint triangular petals. Its direct Sachs calculation proves $s^+ > s^-$, hence $s^+ > |E| = |V| + 3$. This calculation, including the exact matching injections which dominate the three-cycle Sachs term, appears in [7, Proposition 4.11]. Bound (8) is the exact shared C_3, C_5, C_5 phase bound in [4]; (9) is the tetracyclic cactus theorem [5]. These are the only external packet dependencies used below.

2 Cluster trees and exact territories

Join two cyclic blocks in the *shared-cut graph* when they share a cut vertex, and call its connected components shared-cut clusters. In the block-cut tree of G , contract the incidence subtree belonging to each shared-cut cluster to one marked node. Every cluster, including every singleton cyclic block, is marked. Take the minimal subtree spanning all marked nodes and suppress only unmarked degree-two nodes. The resulting tree is the *reduced cluster tree*. A reduced edge therefore represents a path in the block-cut tree whose block nodes are all bridge blocks; it contains no unassigned cyclic block.

Lemma 2.1 (Arbitrary connector territories). *Let $S_1, \dots, S_k$ be pairwise vertex-disjoint connected subtrees of the reduced cluster tree, and suppose that every marked cluster node belongs to exactly one S_i . Then $V(G)$ has a partition into connected induced territories $G_1, \dots, G_k$ such that G_i retains exactly the cyclic blocks whose marked nodes lie in S_i . All cuts between territories occur on actual bridge edges. Every tree attached anywhere to the cyclic hull is assigned whole to one territory.*

Proof. First contract each S_i to a labelled root in the reduced cluster tree. Assign every remaining reduced-tree vertex to a nearest labelled root, using a fixed label order to break ties. In a tree, the

vertices assigned to one root induce a connected subtree: along the unique path toward that root the distance decreases by one, while no competing distance can become strictly smaller after the fixed tie rule. Thus this assignment partitions the reduced tree into connected labelled subtrees. Every edge between two labels is a reduced edge and hence expands to a path consisting only of bridge blocks. Choose one actual bridge on that path as the cut. Assign the two path remnants to their respective labels.

Equivalently, at an unmarked Steiner branch one may assign the branch vertex and one initial segment to one incident territory and cut one actual bridge on every other incident segment. This explicitly includes a genuine Y-shaped connector and does not assume a linear order of clusters. After all selected bridge cuts, each component of the cyclic hull is connected and contains exactly its assigned cyclic blocks.

Finally, a component outside the cyclic hull meets it at a unique vertex. Two attachments would create a cycle in the block-cut tree. Assign the entire hanging component to the territory containing that attachment. The resulting vertex sets induce connected subgraphs and retain precisely the asserted cycles. Edges crossing between vertex sets are discarded only through (1); no edge-monotonicity statement is used. $\square$

We will also split a cyclic block into proper consecutive intervals. If the shared vertices of a cycle C are listed cyclically and the node C is deleted from its cycle-cut incidence tree, the resulting branches can be assigned nonempty consecutive intervals of C , one interval containing the shared vertex of each branch. Every interval induces a proper path, so the split cycle contributes no cycle to any territory. Several cyclically adjacent marks may instead be put in one interval. Hanging trees are assigned by their unique attachment as in Lemma 2.1.

3 The exact DNN residual reduction

For completeness, we record the exact reduction rather than merely citing its output. For a connected edge-containing graph define

$$\kappa(G) = \sup_{\substack{M \succeq 0, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [6, 8] states that, if G has b bridge blocks and cyclic blocks of lengths $\ell_1, \dots, \ell_r$, then

$$\kappa(G) = b + \sum_{j=1}^r \kappa(C_{\ell_j}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}, \end{cases} \quad s^-(G) \leq \kappa(G). \quad (11)$$

Here is the proof mechanism behind the last inequality and the exact additivity. Fold every nonedge entry M_{uv} into the two diagonal entries by adding $M_{uv}(e_u - e_v)(e_u - e_v)^T$. This preserves the denominator and the edge entries. Rotational averaging on C_ℓ , concavity of the square root, and the least Fourier eigenvalue give the two cycle constants in (11); equality is attained by $x(2cI + A(C_\ell))$, where $c = 1$ for even ℓ and $c = \cos(\pi/\ell)$ for odd ℓ .

For a one-vertex sum $G = G_1 \vee_z G_2$, use a folded Gram system and decompose $g_z = u + w + h$, where u and w lie in the mutually orthogonal spans of the two sides and h is orthogonal to both. Use $u + h$ as the cut vector on the first side and w on the second. Both the edge numerator sum and the denominator split; Cauchy-Schwarz gives $\kappa(G) \leq \kappa(G_1) + \kappa(G_2)$. Conversely, glue

near-optimal Gram systems orthogonally and scale them so that $T_1/\kappa(G_1) = T_2/\kappa(G_2)$, proving equality. Finally write the adjacency matrix as $A = A_+ - B$. The Schur product theorem makes $B \circ B$ doubly nonnegative, while

$$s^-/2 = - \sum_{uv \in E(G)} B_{uv} \leq \sum_{uv \in E(G)} |B_{uv}|, \quad \mathbf{1}^T (B \circ B) \mathbf{1} = \text{tr } B^2 = s^-.$$

The definition of κ yields $(s^-)^2 \leq \kappa(G)s^-$ and hence the last assertion of (11).

Put

$$\epsilon_\ell = \begin{cases} 0, & \ell \text{ even,} \\ \ell \tan^2(\pi/(2\ell)), & \ell \text{ odd.} \end{cases}$$

Then

$$\epsilon_3 = 1, \quad \epsilon_5 = 5 - 2\sqrt{5}, \quad \epsilon_5 + \epsilon_7 < 1, \quad (12)$$

and ϵ_ℓ strictly decreases through odd lengths. The monotonicity follows, after putting $u = \pi/(2x)$, from the positivity of the derivative of $\tan^2 u/u$, since $2u > \sin u \cos u$.

For a pentacyclic cactus with cycle lengths $\ell_1, \dots, \ell_5$, block counting gives

$$b + \sum_{i=1}^5 \ell_i = n + 4.$$

Since $s^+ + s^- = 2|E| = 2(n + 4)$, (11) gives the exact residual estimate

$$\sigma(G) = s^+(G) - n \geq 4 - \sum_{i=1}^5 \epsilon_{\ell_i}. \quad (13)$$

If at most two cycles are triangles, then $\sum_i \epsilon_{\ell_i} \leq 2 + 3\epsilon_5 < 4$. With exactly three triangles, (12) shows that the sum can reach 4 only when both remaining cycles are pentagons: replacing either pentagon by an even cycle or an odd cycle of length at least seven makes their combined contribution less than one. With at least four triangles, designate four of them and call the fifth cycle Q ; this includes $Q = T$. Consequently (13) proves $\sigma(G) > 0$ unless the cycle multiset belongs to exactly one of the residual families

$$TTTTQ = \{3, 3, 3, 3, q\} \quad (q \geq 3), \quad TTTTP = \{3, 3, 3, 5, 5\}. \quad (14)$$

4 Disconnected shared-cut graph: the $TTTTQ$ family

Proposition 4.1. *Suppose the five cyclic blocks are $TTTTQ$ and the shared-cut graph is disconnected. Then $s^+(G) > |V(G)|$.*

Proof. If $q \equiv 1 \pmod{4}$, the complete list of proper colored set partitions, up to permutation of the four triangles, consists of the following eleven rows. Each row records an induced-territory certificate. The cluster territories are realized by Lemma 2.1; a displayed zero is nonnegative, while a strict triangular term makes the total strict.

partition	lower surplus certificate
TTTT Q	$> 3 - \delta_q > 0$
TTTQ T	positive tetracyclic territory plus a strict T
TTT TQ	$> 2 + (1 - \delta_q) > 0$
TTQ TT	nonnegative tricyclic territory plus > 1
TTT T Q	$> 2 + 0 - \delta_q > 0$
TT TT Q	$> 2 - \delta_q > 0$
TTQ T T	nonnegative tricyclic territory plus two strict T 's
TT TQ T	$> 1 + (1 - \delta_q) + 0$
TT T T Q	$> 1 + 0 + 0 - \delta_q > 0$
TQ T T T	$> 1 - \delta_q$ plus strict T 's
T T T T Q	reduced-tree leaf argument below

The table uses (2)–(9); in particular the connected shared-cut TTT and $TTTT$ clusters satisfy (5) and (6).

For the final row, inspect the minimal reduced tree spanning the five marked cycle nodes. Every leaf of this minimal tree is marked. If a triangle is a leaf, cut the first actual bridge toward the rest. Its territory has strict positive surplus, and the connected remainder is tetracyclic and has strict positive surplus by (9). If no triangle were a leaf, every leaf would have to be the unique Q node, impossible because a finite nontrivial tree has at least two leaves. This proves the all-singleton row without assuming that the reduced tree is a path.

If q is even, an even unicyclic territory has zero surplus. Apply the same cluster territories: every territory with at most three cycles is nonnegative, every tetracyclic territory is positive, and the proper cluster partition always leaves either a strict triangular territory or a positive tetracyclic territory. The all-singleton leaf proof is unchanged. If $q \equiv 3 \pmod{4}$, every cluster with at most three cycles has packing number at most two and is strict by (10); every four-cycle territory is strict by the tetracyclic theorem (9). Singleton territories are strict as well. Thus these two nonhostile parities are also covered. $\square$

5 Disconnected shared-cut graph: the $TTTPP$ family

The direct cluster ledger leaves two entry-sensitive partitions. We prove them before displaying the complete audit.

Lemma 5.1 (The $TTTP|P$ repair). *Suppose one shared-cut cluster A has cycles T_1, T_2, T_3, P_0 , and a remote pentagon P_1 is bridge-separated from it. Then $s^+(G) > |V(G)|$.*

Proof. If two triangles of A meet, let k be the number of shared cyclic cuts on P_0 . The incidence excess for four cycle nodes is three. The k cuts on P_0 contribute at least k , and the intersecting triangle pair contributes one additional unit, whether its common cut is off P_0 or is a cut of degree at least three on P_0 . Hence $k \leq 2$. Choose a private vertex v of P_0 , put v and all off-cycle branches rooted there in a tree F , and put the rest of A in H . Then $P_0 - v$ connects all of its shared cuts, so H is connected. Its only cycles are the three triangles, including an intersecting pair; hence their packing number is at most two. Thus

$$\sigma(A) \geq \sigma(F) + \sigma(H) > -1 + 2 = 1.$$

Together with $\sigma(P_1) \geq -\delta$, connector territories give total surplus $> 1 - \delta > 0$.

Assume now that the three triangles are pairwise disjoint. Incidence-tree acyclicity forces P_0 to be central, with T_i attached at three distinct vertices x_i . Let the connector from P_1 first meet the cyclic core of A at z .

If it enters through T_i , including at x_i , put all of T_i , the connector, and P_1 in one territory. Put $P_0 - x_i$ and the other two triangles in the other. These territories have cycle multisets TP and TT . An entry through a branch rooted on T_i is the same case.

If the connector enters P_0 at a private vertex z , cyclically order z, x_1, x_2, x_3 . Choose an x_i adjacent to z in this cyclic order and cut the two boundary edges of a consecutive interval containing exactly the pair z, x_i among these four marks. This interval, T_i , the connector, and P_1 form a TP territory. The complementary proper interval and the other two triangles form a TT territory. All branches rooted at a boundary vertex go with the territory owning that vertex.

In every entry topology, the surplus is

$$> (1 - \delta) + 1 > 0.$$

The split of P_0 is essential: it prevents the common attachment vertex from being assigned to two packets. $\square$

Lemma 5.2 (The $TTP|T|P$ repair). *Suppose one shared-cut cluster A has cycles T_1, T_2, P_0 , while the singleton clusters are $B = T_3$ and $C = P_1$. Then $s^+(G) > |V(G)|$.*

Proof. If T_1, T_2 share a cut, (7) gives $\sigma(A) > 2 - \delta$; with $\sigma(T_3) > 0$ and $\sigma(P_1) \geq -\delta$, the total is $> 2 - 2\delta > 0$. Thus assume that the incidence in A is the distinct-cut chain

$$T_1 - P_0 - T_2,$$

where T_i meets P_0 at x_i .

Use the actual minimal connector tree on the three marked clusters. If the $B-C$ path avoids A , take the entire $B-C$ connector subtree as a TP territory and A as the other territory. Their surplus is $> 1 - \delta + 0$. This includes a genuine Steiner Y : assign its branch vertex to the $B-C$ territory and cut the A branch on an actual bridge.

Suppose instead that the $B-C$ path passes through A , so there are two external connector entries into its cyclic core. An entry through T_i , including at x_i , is represented by a mark at x_i forced to travel with T_i ; an entry at a private vertex of P_0 is marked at that vertex.

If both entries are forced to the same T_i , put both external cycles, both connector branches, and all of T_i in one TTP territory. Put $P_0 - x_i$ and the other triangle in a strict triangular territory. This also covers coincident entries at x_i .

Suppose the external pentagon entry is forced to T_i , but the external triangle entry is not. Isolate x_i by two cuts of P_0 . Its interval carries $P_1 + T_i$ and is a TP territory; the complementary interval carries $T_3 + T_j$ and is a TT territory. If instead the external triangle entry is forced to T_i , isolate x_i on the TT side and put $P_1 + T_j$ on the complementary TP side.

It remains that both entries lie at private vertices of P_0 . If they coincide, take a proper interval containing that common mark and one adjacent x_i . It gives a TTP territory, while the complement gives a triangular territory. If the entries are distinct, cyclically order the two labelled external marks and the two unlabeled marks x_1, x_2 . One neighbor of the pentagon-entry mark is an x_i such that the other x_j is adjacent to the triangle-entry mark; this follows by checking the three cyclic orders of two labelled and two unlabeled marks. Cut in the two gaps between these adjacent pairs. The resulting proper intervals carry $P_1 + T_i$ and $T_3 + T_j$, hence give TP and TT territories.

Every case therefore gives either

$$TP + TT, \quad \sigma > (1 - \delta) + 1 > 0,$$

or

$$TTP + T, \quad \sigma > 0,$$

where the tricyclic packet is nonnegative and the triangle is strict. This covers private cycle entries, shared-cut entries, entries through attached triangles, coincident roots, and branching connectors. $\square$

Proposition 5.3. *Suppose the five cyclic blocks are $TTTP$ and the shared-cut graph is disconnected. Then $s^+(G) > |V(G)|$.*

Proof. Up to permutation of equal cycles, the complete list of fifteen proper colored set partitions is the following. The first and seventh rows invoke Lemmas 5.1 and 5.2; all other non-singleton rows use the direct packet ledger.

partition	certificate or lower surplus
$TTTP P$	Lemma 5.1
$TTT PP$	$> 2 + 0$
$TPPP T$	positive tetracyclic territory plus a strict T
$TTP TP$	$> 0 + (1 - \delta)$
$TPP TT$	$> (6 - 2\sqrt{5}) + 1$
$TTT P P$	$> 2 - 2\delta$
$TTP T P$	Lemma 5.2
$TT TP P$	$> 1 + (1 - \delta) - \delta$
$TT T PP$	$> 1 + 0 + 0$
$TPP T T$	$> 6 - 2\sqrt{5}$ plus strict T 's
$TP TP T$	$> 2(1 - \delta)$ plus a strict T
$TT T P P$	$> 1 - 2\delta$ plus a strict T
$TP T T P$	$> 1 - 2\delta$ plus strict T 's
$PP T T T$	nonnegative PP plus strict T 's
$T T T P P$	reduced-tree leaf argument below

Every displayed quantity is positive because $\delta < 1/2$.

For the all-singleton row, inspect the minimal reduced tree spanning the five marked cycle nodes. Every leaf is marked. If a triangle is a leaf, cut its first actual bridge; the triangular territory is strict, and the connected tetracyclic remainder is strict by (9). If no triangle is a leaf, all leaves are among the two pentagons. There must be at least two leaves, so both pentagons are leaves and there are exactly two leaves. A finite tree with exactly two leaves is a path, with all three triangle marks internal. Cut actual bridges so that the path is partitioned into TP , T , and TP territories. Their total surplus is $> 2(1 - \delta) > 0$. This argument remains valid when reduced edges expand to long paths or pass through previously suppressed degree-two vertices. $\square$

6 Fully shared $TTTTQ$

For one shared-cut cluster, let I be the bipartite incidence tree whose cycle nodes are the five cyclic blocks and whose cut nodes are the vertices belonging to at least two cyclic blocks. If there are c cut nodes, then

$$|E(I)| = c + 4, \quad \sum_{x \text{ cut node}} (\deg_I(x) - 1) = 4. \quad (15)$$

A multiway cut incident with d cyclic blocks contributes exactly $d - 1$.

Proposition 6.1 (Fully shared $TTTTQ$). *Suppose the five blocks $T_1, T_2, T_3, T_4, Q = C_q$, $q \geq 3$, form one shared-cut cluster. Then $s^+(G) > |V(G)|$.*

Proof. First suppose the four designated triangles are pairwise vertex-disjoint. Every cut node incident with a triangle must also be incident with Q , since it has degree at least two and cannot be incident with another triangle. A triangle cannot meet Q at two cuts, because the two triangle–cut– Q routes would form a cycle in I . Connectedness forces each triangle to meet Q exactly once, and the four attachment cuts are distinct. Thus Q is central and the four triangles are pairwise disjoint petals. This forces $q \geq 4$.

List the four attachments in cyclic order and partition $V(Q)$ into four nonempty consecutive cyclic intervals, one containing each attachment. Choose one boundary edge in each arc between consecutive attachments. Every interval is a proper path, including when attachments are adjacent; when $q = 4$, all four intervals are singletons. Adjoin to each interval its triangular petal and assign every off-core tree by its unique core attachment. This partitions G into four connected induced triangular unicyclic cacti, each of strict positive surplus by (2). Equation (1) proves the result.

Now suppose some two designated triangles meet. Let k be the number of shared cyclic cuts on Q . Those cuts contribute at least k to (15). The intersecting pair contributes one additional unit: its common cut is either an additional cut off Q , or a cut on Q of degree at least three and hence contributes at least two. Therefore $k + 1 \leq 4$, so $k \leq 3$.

Assume $q \geq 4$. Choose a vertex v of Q that is not a shared cyclic cut. Let F consist of v and every component of $G - E(Q)$ attached to Q only at v , and put $H = G - F$. Then F is a tree and

$$s^+(F) = |F| - 1. \quad (16)$$

The path $Q - v$ contains every shared cut on Q ; replacing the Q node of I by this path leaves all four triangular blocks connected. Thus H is connected and induced, and its only cyclic blocks are the four triangles.

Partition H along bridge connectors between its shared-cut clusters using Lemma 2.1. A cluster containing exactly two intersecting triangles has surplus > 1 . One containing three triangles has packing at most two and surplus > 2 . A four-triangle cluster has surplus > 3 : use (10) when its packing number is at most two and (6) for the central-triangle/three-petal incidence when the packing number is three. Every remaining singleton triangular packet has strict positive surplus. Hence the cluster containing the chosen intersecting pair supplies more than one unit and all others supply positive surplus, so

$$s^+(H) > |H| + 1.$$

Together with (16) and (1), this gives

$$s^+(G) \geq s^+(H) + s^+(F) > |H| + 1 + |F| - 1 = |V(G)|.$$

Finally, if $q = 3$, all five cyclic blocks are triangles and the designation of Q is arbitrary. The universal triangular block-graph theorem [7, Corollary 4.2] applies directly and gives $s^+(G) > |V(G)|$, including all multiway cuts. The four-petal alternative cannot have a triangular center because it requires four distinct attachment vertices. These alternatives exhaust every incidence. $\square$

7 Fully shared $TTTPP$: exact incidence census

Continue to use the incidence tree I and (15). Every cut node has degree at least two, so $1 \leq c \leq 4$. Distinct incidences use distinct vertices of a cycle; hence a triangle cycle node has degree at most three and a pentagon cycle node degree at most five.

If an internal cycle node is split into consecutive intervals, deleting it from I gives one branch multiset for each interval. Encode such a multiset by (t, p) . Splitting a triangle leaves total multiset $(2, 2)$, and the following are exactly the positive branch partition types used by the census:

branches after splitting T	lower surplus
$PP + TT$	> 1
$T + TPP$	$> 3/2$
$TP + TP$	$> 2 - 2\delta$
$P + P + TT$	$> 1 - 2\delta$
$P + T + TP$	$> 1 - 2\delta$
$PP + T + T$	> 0

Splitting a pentagon leaves total multiset $(3, 1)$, with positive types

branches after splitting P	lower surplus
$P + TTT$	$> 2 - \delta$
$T + TTP$	> 0
$TP + TT$	$> 2 - \delta$
$P + T + TT$	$> 1 - \delta$
$T + T + TP$	$> 1 - \delta$

Here a strict zero means a strict triangular packet is added to a nonnegative packet. Every bound follows from (2)–(8), and all are positive because $\delta < 1/2$.

Lemma 7.1 (Exact colored incidence census). *Modulo permutations of the three triangle nodes, the two pentagon nodes, and the cut nodes, the numbers of admissible colored incidence trees for $c = 1, 2, 3, 4$ are respectively*

$$1, 7, 18, 14. \tag{17}$$

Of these, a one-cycle split certified by one of the two preceding tables resolves respectively

$$0, 6, 17, 13. \tag{18}$$

The four unresolved colored trees are exactly:

1. one cut x incident with all five cycles;
2. cuts of degrees $(4, 2)$, where x meets T_1, T_2, T_3, P_1 and y meets T_1, P_2 ;
3. cuts of degrees $(3, 2, 2)$, where x meets T_1, T_2, T_3 , y meets T_1, P_1 , and z meets T_2, P_2 ;
4. a pentagon hub: P_1 meets T_1, T_2, T_3, P_2 at four distinct degree-two cuts.

Proof. The assertion is a finite exact enumeration. From the repository root it is reproduced by

`python research/pentacyclic-ttttp-incidence-census.py.`

For each $c \in \{1, 2, 3, 4\}$ it enumerates subsets of the $5c$ possible cycle–cut edges having $c + 4$ edges. It rejects a subset unless every cycle degree is positive, every cut degree is at least two, triangle degrees are at most three, pentagon degrees are at most five, and union–find verifies the tree condition.

It canonicalizes each surviving edge set over all permutations of the three triangle labels, both pentagon labels, and all cut labels. For every cycle node it then deletes that node, computes the colored branch components by depth-first search, and compares their sorted (t, p) tuple with the exact sets

$$\begin{aligned}\mathcal{G}_T &= \{((0, 2), (2, 0)), ((1, 0), (1, 2)), ((1, 1), (1, 1)), \\ &\quad ((0, 1), (0, 1), (2, 0)), ((0, 1), (1, 0), (1, 1)), ((0, 2), (1, 0), (1, 0))\}, \\ \mathcal{G}_P &= \{((0, 1), (3, 0)), ((1, 0), (2, 1)), ((1, 1), (2, 0)), \\ &\quad ((0, 1), (1, 0), (2, 0)), ((1, 0), (1, 0), (1, 1))\}.\end{aligned}$$

These are precisely the two displayed packet tables. The script asserts the counts (17), the resolved counts (18), and the following canonical unresolved edge sets, where cycle nodes 0, 1, 2 are triangles, 3, 4 are pentagons, and cut nodes start at 5:

$$\begin{aligned}c = 1 : & \quad ((0, 5), (1, 5), (2, 5), (3, 5), (4, 5)); \\ c = 2 : & \quad ((0, 5), (0, 6), (1, 5), (2, 5), (3, 5), (4, 6)); \\ c = 3 : & \quad ((0, 5), (0, 6), (1, 5), (1, 7), (2, 5), (3, 6), (4, 7)); \\ c = 4 : & \quad ((0, 5), (1, 6), (2, 7), (3, 5), (3, 6), (3, 7), (3, 8), (4, 8)).\end{aligned}$$

These translate exactly into the four listed incidences. Thus the census is both color-preserving and exhaustive. $\square$

Proposition 7.2 (Fully shared *TTTTP*). *Suppose the five blocks T_1, T_2, T_3, P_1, P_2 form one shared-cut cluster. Then $s^+(G) > |V(G)|$.*

Proof. For each of the thirty-six incidence trees resolved by Lemma 7.1, split the certified cycle into proper consecutive intervals, assign one interval to each incidence-tree branch, and assign every off-core tree by its unique attachment. The resulting connected induced territories have exactly the branch multisets in the corresponding packet table, whose total surplus is positive. Equation (1) settles all thirty-six.

In exceptions 1–3, both pentagons are leaf cycle nodes and the three triangles remain a connected shared-cut cluster after the pentagons are opened. Choose a private vertex v_i on each P_i . Let F_i contain v_i and every off-cycle tree branch rooted there, and put all remaining vertices in H . Each F_i is a tree, so $\sigma(F_i) = -1$. Each path $P_i - v_i$ contains the unique shared cut of its leaf pentagon, so H is connected. Its only cycles are the three triangles, which form one connected shared-cut cluster. At least two meet, the packing number is at most two, and (5) gives $\sigma(H) > 2$. Therefore

$$\sigma(G) \geq \sigma(H) + \sigma(F_1) + \sigma(F_2) > 2 - 1 - 1 = 0.$$

In exception 4, cyclically order the four marked vertices on the hub P_1 . The mark belonging to P_2 has a cyclic neighbor belonging to some triangle, say T_1 . Split P_1 into three proper consecutive intervals: one contains those two neighboring marks, and each of the other two contains the mark of T_2 or T_3 . The induced territories have types TP , T , and T . Consequently

$$\sigma(G) > (1 - \delta) + 0 + 0 > 0.$$

All forty colored incidence trees are covered. $\square$

8 Main theorem

Theorem 8.1. *Every connected pentacyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The exact DNN estimate (13) proves the assertion unless the cycle multiset is $TTTTQ$ or $TTTTPP$. If the shared-cut graph is disconnected, Propositions 4.1 and 5.3 settle the two residual families, using the complete eleven-row and fifteen-row colored partition audits. If it is connected, Propositions 6.1 and 7.2 apply. These alternatives exhaust all cycle lengths and all incidences. Lemma 2.1 and every interval split explicitly allow arbitrary connector trees and arbitrary attached trees. Every spectral comparison is applied to vertex-disjoint induced subgraphs; neither edge monotonicity nor tree relocation is used. $\square$

Since a connected pentacyclic graph has $|E(G)| = n + 4$, the theorem proves the strict conclusion of the Akbari–Kumar–Mohar–Pragada–Zhang conjecture [2, Conjecture 1.2] for every pentacyclic cactus.

A Exact reproduction and local proof objects

From the repository root, run

```
python research/pentacyclic-ttttp-incidence-census.py
```

The expected output is

```
colored trees by cut count: {1: 1, 2: 7, 3: 18, 4: 14}
one-cycle split exceptions:
c=1: ((0, 5), (1, 5), (2, 5), (3, 5), (4, 5))
c=2: ((0, 5), (0, 6), (1, 5), (2, 5), (3, 5), (4, 6))
c=3: ((0, 5), (0, 6), (1, 5), (1, 7), (2, 5), (3, 6), (4, 7))
c=4: ((0, 5), (1, 6), (2, 7), (3, 5), (3, 6), (3, 7),
      (3, 8), (4, 8))
```

The script contains internal assertions for these counts, for the resolved counts $\{2 : 6, 3 : 17, 4 : 13\}$, and for the exact four exception tuples. The local manuscripts supplying the stated dependencies are

```
sharp-cactus-dnn/paper.tex
all-bicyclic-cacti/paper.tex
all-tricyclic-cacti/paper.tex
all-tetracyclic-cacti/paper.tex
packing-two-square-energy/paper.tex
```

The audited source proof objects integrated into the present manuscript are

```
research/pentacyclic-cactus-complete-audit-2026-07-26.md
research/pentacyclic-fully-shared-3333q-2026-07-26.md
research/pentacyclic-ttttp-fully-shared-2026-07-26.md
research/pentacyclic-ttttp-incidence-census.py
```

References

- [1] S. Akbari, H. Kumar, B. Mohar, and S. Pragada, *A linear lower bound for the square energy of graphs*, Electron. J. Combin. 32(3) (2025), Paper No. P3.53.
- [2] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.
- [3] Companion manuscript, *Positive Square Energy of Every Bicyclic Cactus*, 2026; repository path `all-bicyclic-cacti/paper.tex`.
- [4] Companion manuscript, *Positive Square Energy of Every Tricyclic Cactus*, 2026; repository path `all-tricyclic-cacti/paper.tex`.
- [5] Companion manuscript, *Positive Square Energy of Every Tetracyclic Cactus*, 2026; repository path `all-tetracyclic-cacti/paper.tex`.
- [6] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; repository path `sharp-cactus-dnn/paper.tex`.
- [7] Companion manuscript, *Square Energy from Cycle Packing and Block-Graph Territories*, 2026; repository path `packing-two-square-energy/paper.tex`.
- [8] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.